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Review article

# A Survey of Indoor Location Technologies, Techniques and Applications in Industry

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A B S T R A C T

The recent academic research surrounding indoor positioning systems (IPS) and indoor location-based services (ILBS) are reviewed to establish the current state-of-the-art for IPS and ILBS. This review is focused on the use of IPS / ILBS for cyber-physical systems to support secure and safe asset management (including people as assets), exploring the potential applications of IPS for industry as suggested in the literature. Current application areas in industry are presented, separated into physical item and human traceability applications for context. The literature are reviewed to identify gaps in the ILBS development for industrial applications, future research needs to focus a development framework to enable scalable solutions for industry. The key gaps identified in the literature are: (i) a lack of pathways to extend IPS research into an ILBS, (ii) no end-to-end ILBS have been developed and (iii) no framework has been reported that outlines the information pathways from sensor data collection and location information to an established ILBS. The technologies reviewed are presented in a comparison table (Table 1) intended as a reference for selecting technologies for future systems based on requirements. The techniques used to extract location information from each of the technologies identified are also explored stating current accuracy and aligning the techniques with their suitable technologies.

## 1. Introduction

Cyber-physical systems (CPSs) are one of the key enablers for the Industry 4.0 and Industry 5.0 paradigms for the realisation of dynamic, sustainable and adaptable manufacturing [1]. True CPSs utilise multi-directional links between physical phenomena in the real world and appropriate cyber representations of those phenomena through the capture and processing of data from smart devices and the Industrial Internet of Things (IIoT) [2–4]. This is to be contrasted with “digital shadows” that only support the depiction of data within the cyber-based twin and do not interact with the physical system [5].

The introduction of accurate location information in a CPS is vital to support context aware applications (e.g. asset routing and security), particularly where users of the IIoT are mobile and their locations and paths are not fixed [6,7]. Mobile assets can include human and non-human in warehousing, assembly, logistics and production. Supply chain managers have identified supply chain visibility as a global challenge with significant potential for improved performance [8]. For example, a lack of visibility in the supply chain may leave it vulnerable to asset theft or loss, of particular importance for electrical and electronic goods that could contain personal information covered by the general data protection regulation (GDPR) and potential fines for breaches [8]. In addition to the industry benefits, clients are becoming more insistent on better visibility of their assets in transit and, for specific goods, regulators are

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requiring confirmation of asset environmental integrity in transit (e.g. maintained temperature and humidity) particularly within the pharmaceutical, medical and fresh food domains [8–11].

In scenarios where humans interact with the CPS it is important that contextually relevant services and information are displayed to the user to support appropriate behaviour and informed action [12]. For the provision of appropriate information, a system with context awareness is required and, in the case of factory environments, current user locations can provide key contexts for communication of information that may be pertinent to their task, safety or security [12]. The provision of contextually relevant information within CPS is fundamentally dependent on accurate sensors and knowledge of the location of both fixed and mobile assets.

Although outdoor location services are a mature and robust field, (i.e. the Global Navigation Satellite System (GNSS) is used for context-based information such as navigation and maps), indoor location services remain an open area of research with a variety of technologies available to provide data [13–15]. The scale of the research in this area is evident, with over 2600 journal articles published in the last 5 years relating to "indoor positioning system" or "indoor location". Properties of signals from various technologies can be used to determine location measurements and a wide range of techniques are available to develop an IPS [16–26]. The technologies to develop IPSs can be grouped into four categories: (i) *inertial systems*, which utilise self-contained sensors that measure user motion to estimate relative changes in position, (ii) *radio frequency systems,* which utilise technologies that operate in the radio or microwave frequencies of the electromagnetic spectrum, (iii) *acoustic systems,* which create and receive sound waves or reflected sound waves and (iv) *vision systems*, which use either laser ranging or cameras. *Hybrid systems* utilise a combination of technologies from these categories [22].

For each of the four fundamental technologies (i.e. the data collection methods), several techniques (i.e. the analysis methods) have been identified to estimate location based on the data and to overcome limitations (e.g. environmental impact) of the technology. A technology that has not been reviewed in this article is the use of cellular networks. At the time of writing 5G networks were not considered as candidates for indoor positioning systems due to a lack of availability and coverage. However, with increased deployment of 5G, which operates using wideband frequency, emerging solutions may make economical use of cellular networks [22]. Each technology has limitations and suitability criteria that have not previously been cohesively assessed nor analysed.

Several research gaps have been identified in a narrative review of the literature. Firstly, no standardised set of requirements for an accurate and reliable IPS have been reported and research is needed to identify which sensor, or combination of sensors, is/are required to satisfy requirements of an ILBS. The research reported in this paper was conducted to identify key application areas in the literature in addition to exploring the benefits and limitations of the proposed technologies by considering their suitability for use in ILBS to support context aware CPS in Industry. The paper is structured as follows: Potential industrial applications of IPS are discussed regarding assets and personnel in Sections 2 and 3 respectively. The categories of technology are defined and discussed in detail in Section 4 with the techniques used to analyse the technology presented in Section 5. A summary and gaps in the literature are discussed in Section 6.

### *1.1. Motivation and novelty*

The motivation for this work stems from a lack of research aiming to address technology selection for ILBS development regarding infrastructure impact, cost, security and practicality, the paper aims to summarise current state of the art technologies and techniques to support development of such services. There exists a gap in the literature to enable practitioners to easily select the preferred technology and associated technique to achieve an accuracy aligned with their service goal. This selection process is importance to minimise cost implications of testing multiple technologies or over specifying a technology with an unnecessarily high accuracy for a given task. The paper presents a table and recommendations of use case scenarios for each technology. The goal of this is to support ILBS developing practitioners, avoiding unnecessary delays in the ILBS development process due to reselection of technologies and technique. The novelty in this work lies in the wide range of journals explored and specific focus on the implications of adopting each technology and practicality of the IPS for services extending from IPS rather than accuracy only.

## 2. Asset tracking industrial applications and the importance of IPS

Brik et al. propose an efficient and flexible rescheduling process based on the localisation of assets to handle system disruption by considering resource location, accuracy and proximity [27]. In the system, resources are separated by targeting tasks to minimise delays and maximise execution accuracy, achieving "near optimal" scheduling and reducing the time of work by over 30% [27]. A limitation of this study is the purely theoretical analysis conducted (i.e. using only simulation) with no real-world testing or evaluation to identify realistic schedule improvements or practical limitations. Beliatis et al. assessed several technologies for asset tracking and management of processes in a company which produces and manufactures metal housings with electronics against customer requirements [28]. The system incorporates manual processes, including data input to the enterprise resource planning system [28]. The purpose of the system is for the company to query and track location of assets, their path, up-time and down-time of work in progress (WIP) in order to improve asset utilisation and production performance [28]. The selected technologies for comparison are passive RFID and a low power wireless area network (LP-WAN) system named Sigfox. The Sigfox system is the more expensive solution (€2 per chipset plus €1 per year, each Sigfox base station costs > €4000 each) but had a much larger coverage range (up to 13 km) whilst radio frequency identification (RFID) is the lower cost (10 cents per tag plus reader with no cost estimation given) local solution (up to 10 m) [28]. They highlighted the importance of sufficient support for such a system to help employees in the transition to a digitalised approach, however, no environmental sensing is considered to ensure asset integrity during processing. The solution proposed with Sigfox provided continual asset location tracking for assets at high risk of loss. However, although an industrial use case was used for the prototype testing, the authors did not propose how such a system would be implemented or evaluate the real-world efficacy of the system.

Kumar et al. utilise a combination of passive ultra high frequency (UHF) RFID and Bluetooth low energy (BLE) components to track raw materials and finished goods within a manufacturing facility [29]. The aim was to aid information flow regarding product and processes, increasing supply chain transparency [29]. A lack of asset tracking information has been indicated as an issue in most manufacturing firms, confirming that asset tracking is necessary to avoid mismatched inventories and to improve supply and demand [29]. The system stores the relevant material information on a BLE device attached to each asset. The BLE data include material details, entry location, quantity and relevant documents associated with the asset [29]. The benefits of this system are reported as better visibility of stock in the manufacturing process and tracking number of completed products. However, the system is limited as no environmental information is recorded and only coarse location information is used, identified only when raw material is entered into the system and when a product exits the system. Information regarding processing steps or asset location(s) within the manufacturing site is not included, leaving the system vulnerable to missed production steps. An asset management system with improved granularity is presented by Varshney et al. (2020). They utilise BLE beacon received signal strength (RSS) for room level classification using a support vector machine (SVM) to achieve room location accuracy of 85% - 90%, though no specific industry was targeted the system has been designed to track larger assets, easily located by the user within a room [30]. The benefit to this system is that the radio usage requirements are minimal (i.e. 4 min broadcasting at 10 Hz every hour), extending the battery life of the BLE beacons to over a year (417 days), with further power savings indicated through the use of an inertial wake-up sensor [30]. The limitations of this system are that it was only tested in small rooms, where a single beacon is sufficient to cover the room so scalability to large scale manufacturing and business enterprises will be an issue. The system lacks in-depth asset information and finally, as with [29], the granularity of the system is limited to room location only.

### 2.1. Smart containers

Smart containers have been linked with the premise of the Physical Internet (PI), a global logistics system which aims to move, handle, store, and transport assets efficiently and sustainably [31]. Although the authors in [32] discuss the importance and requirements for a PI system for logistics, no implementation solutions are proposed. Towards an implementation of the PI paradigm, Falkenberg et al. present a warehouse testbed, the Physical Internet Laboratory (*PhyNetLab*) for prototyping smart containers, retrofitted with a *PhyNode* in a realistic scenario [33]. The *PhyNodes*, described in more detail in [34] are an IoT device which is attached to a container designed to be energy neutral through an energy harvesting photovoltaic cell. The *PhyNode* is equipped with multiple sensors to enable the system to collect ambient information about the asset they are attached to in addition to ensuring optimal photovoltaic cell operation [34]. The system was tested for scalability by operating 38 *PhyNodes* simultaneously, however due to communications packet collisions this increases average energy consumption by more than double (57 to 123 mJ) [33] making it less suitable for industrial applications. Although the sensors for environmental and shock loading conditions are available to the *PhyNode*, they are currently not utilised in the *PhyNetLab* testing procedure and the system has only been tested in a lab modelling an industrial environment not subject to transit, where environmental conditions may be continually changing. Korotky et al. present a smart container comprising a GPS enabled container and an actuated mechanical frame that could be flattened (when not in use) to allow multiple containers to be stacked, reducing storage required for empty containers [35]. The system was supported by an application developed to optimise container delivery routing to reduced fuel consumption and container collection time [35]. Unfortunately, as with other systems reported, the system yields no additional information about the environmental conditions or handling of the assets being transported. Another solution for a smart container, though not directly aimed at implementation of PI, is the smart Returnable Transit Item (smaRTI) presented by Neal et al. [36]. The solution is a sensor enabled pallet for transporting engine blocks within and between manufacturing facilities [36]. The solution proposed is controlled by a 32-bit microcontroller and utilises RFID to allow unit identification. Alongside this, several sensors including an accelerometer, gyroscope, magnetometer, altimeter, temperature and humidity sensors allow data collection about the asset environment and detection of motion, including shock loading [36]. The solution proposed in [36] is comprehensive and has been tested in a real-world manufacturing environment. Because this implementation was tested in real world use, several limitations were identified by the authors. The limitations of the system were the waypoint nature of its location information, i.e. the RFID is used to detect location at specific points about a site. Additionally, the implementation is focused on the hardware and management level monitoring of process for logistics. However, the smaRTI has been implemented in a static location only (e.g. at site entry/exit) and not tested for use during transport to determine asset path information. A commonality found in the literature reviewed, is that few researchers have considered a realistic scenario or specific industry in which the asset tracking could be utilised. Furthermore, implementation and viable user interfaces for the asset tracking systems which would portray useful information to the user have not been explored.

### 2.2. Asset integrity certification

Hasan and Salah developed a blockchain based proof of delivery process that initiates an authorised, auditable, time-bound and most importantly immutable contract for transactions involving assets between two parties [37]. One limitation of the process is that it requires a human to confirm each process step, with no guarantee of asset presence and the system only logs that the asset has reached its destination. Hasan et al. discuss current work on how blockchain may be leveraged to support the advancement of PI services, outlining a framework for a blockchain system within which a PI service could be built [38]. The primary reported benefit of using blockchain is that trust is enhanced and the opportunities for fraud reduced due to the immutability of the distributed shared ledger.

However, a key limitation is raised due to the lack of research into scalability of blockchain supported services [38]. No work has been reported in the literature whereby a working blockchain has been implemented for PI purposes. If successful, such an approach could influence contractual requirements and dictate that the delivery is performed with specified integrity requirements, resulting in immutable delivery quality, thereby reducing waste (lost or damaged assets) and improving trust.

## 3. Personnel tracking applications

Employers have a legal responsibility to provide and maintain safe working environments [39]. The requirement for safe working includes (as far as is reasonably practicable): (i) the development of safe and well-maintained workspaces, (ii) appropriate and up to date safe working practices, (iii) information dissemination & training and (iv) the provision of suitable protective equipment and clothing [39]. Despite efforts by employers to support safe working practices, there remains a substantial number of fatal and non-fatal injuries occurring at work sites [40]. The UK statistics (2019/20) indicate that 111 workers were killed during working hours with 40 of these fatalities in the construction and 15 in manufacturing sectors [40]. The health and safety executive (HSE) reported that 38.3 million working days were lost due to work related illness and injury, 6.3 million lost due to non-fatal workplace injuries, with an average of 9.1 working days lost per person injured at work [41]. The risk of eye injury, hand injury and hearing impairment, could be lowered by 60% or more when the correct Personal Protective Equipment (PPE) is worn, for example, 7.5% of total injuries could be reduced by wearing a helmet and 98% of workers are reported as witnessing others not wearing the appropriate PPE with 30% stating it is a regular occurrence [42]. These data suggest that the primary issue with PPE-based protection policies arise from improper use of PPE. Despite compliance with the requirements and legislation guiding the use of PPE and implementation of appropriate training, the present methods adopted in industry appear to be ineffective at preventing injury and ensuring PPE and safety behaviour compliance.

### *3.1. Detection of PPE*

Kelm et al. develop an RFID gateway for a construction site that would identify PPE of the users entering an area, with RFID tags located on each item of PPE [43]. Their system uses a fingerprint scanner in addition to near-field communication (NFC) tags for identification. The use of fingerprinting may lead to contractual issues regarding General Data Protection Regulation (GDPR), owing to the increased security required to store a user's fingerprint and consent requirements for visitors or contractors [43]. Their solution only determines the presence of PPE at entry/exit. A significant limitation of this approach is that the system is only able to detect the presence of one set of PPE so could not determine if the user would be working in an area which required additional or less PPE. An improved CPS augmented PPE solution could be designed to detect ongoing use of PPE in addition to detecting its presence, for example Zhang et al. detected non-compliance for hard hat use in the construction industry in real-time [44]. The systems proposed consist of a 125 kHz RF signal for communication, triggered by RFID with infrared and thermal sensors, to ensure the hard hat is being worn [44]. A limitation of this solution is that a user needs to login to each piece of equipment using an application and quick response (QR) code. The main issue with this system is that it has sensors physically mounted on the inside of the hard hat. The researchers conduct no testing as to the impact of this on the integrity of the hard hat with regards to comfort or functionality. In the worst-case scenario this could render the hard hat ineffective at preventing injury, so further work is required. Prandana et al. proposed a vision system for detecting correct use of head worn PPE (i.e. hard hats, face mask, earmuffs and safety glasses) [45]. The system is designed so a user could scan an RFID card for identification and the system would take an image, convert this to greyscale and use a convolutional neural network (CNN) to identify the presence of correct PPE. The system is reported to have an 85.83% accuracy for detecting PPE, falsely identifying 18 of 120 tests. A similar system using CNN was presented by Wu et al. for detecting worn hard hats (in addition to colour) in the construction industry, reporting CNN accuracy of approximately 75% [46]. Both systems are limited by their recall accuracies leads to false negatives in which the user is wearing PPE the system does not recognise or, more concerningly, false positives in which users are allowed access to an area with improper PPE. An additional issue with this system, is the impact of using vision systems with regards to GDPR and user acceptance, the research in [45] does not consider user privacy.

### *3.2. Unergonomic postures*

Ranavolo et al. explore the use of wearable IMUs as a form of human activity recognition (HAR) to collect data on body posture in a manual handling task (i.e. lifting) with various lifting energy consumption levels [47]. The system limitations are that no composite or sequential motions were evaluated, therefore deviations from the testing protocols may not be accurately identified. In addition, the testing has been only conducted on males and gender aspects would need to be considered in future work. The use of HAR and wearable sensors has an advantage over traditional static data collection (e.g. ergonomic observation) because it can be collected over longer durations and used to identify repetitive tasks or change in task performance over time. For example Wang et al. collect step counts and accelerometer information from smart devices (i.e. a smartwatch and a smartphone) to supplement self-surveys of physical activity in estimation of lower limb loading [48]. It was concluded in [48], smart devices are a viable method for monitoring workplace physical activity and limb loading.

### *3.3. Trips, slips, falls, working at height*

Current research that could be utilised to identify potential hazards at work is predominantly focused on detecting potential injury events in the elderly or infirm [49]. There is limited use of this approach for industrial settings. An IMU has been used in [50] to detect slips trips and falls, allowing medical attention to be brought to the aid of a worker [51,52]. These technologies could benefit the manufacturing and industrial sectors if used to detect "same level fall'' and "fall to lower level'' events. Campero-Jurado et al. (2020) have developed a smart helmet augmented with sensors with the goal of reducing incidents which lead to injury [53]. This system aims to mitigate the hazards which lead to incidents identifying: (i) lack of adequate lighting, (ii) extreme temperatures, (iii) harmful air quality and (iv) operator movement (impacts) as the key hazards [53]. The system detects shock via a force sensitive resistor and falls via an accelerometer. It also utilises a light sensor to detect darker areas to turn on a helmet mounted light automatically [53]. There are additional patents for similar systems mentioned in their further work [53]. Colombo et al. presents a vision system utilising 2D RGB cameras (i.e. security cameras) to extract a skeleton of a human in the image. A long short-term memory (LSTM) neural network is used to detect falls [54]. This system is limited by use of a camera which raises GDPR concerns and necessitates line-of-sight (LOS) t to operate accurately, which may not be viable in a manufacturing site and requires the infrastructure to be modified to support power, communications and maintenance networked camera systems to cover all blind spots. Manivannan et al. have reviewed the literature on use of barometers for HAR tracking applications and highlight climbing stairs and riding elevators as applications utilising barometers in the literature, although few studies have considered all aspects of daily living [55]. It is also noted in [55] that very few studies use a barometer as a sole sensor with their use restricted to determine altitudes at floor level granularities rather than the finer resolution which would be required for working at heights around manufacturing machines and construction sites. Research exploring this approach has reported that the device provides only a weak indication of context when used to identify motion in an elevator or escalator [56].

### 3.4. *Summary of human traceability opportunities*

The beneficial opportunities to companies and personnel for applying monitoring techniques to human resources primarily include: (i) efficiency (e.g. resource utilisation and location services) (ii) improved security (protection for both the organisation and the individual) (iii) improved health and safety and (iv) efficiency through navigation services, see Fig. 1 collated from [13,22,57–59].

## 4. Technology for IPS

The following commonly used technologies: (i) *inertial systems,* (ii) *radio frequency systems*, (iii) *acoustic systems* and (iv) *vision systems* can be used individually or combined in hybrid systems are examined in the following sections to highlight their limitations and capabilities for use in ILBS.

### 4.1. *Inertial systems*

Inertial measurement units (IMUs) can be used to determine positions relative to a known point [60]. An IMU uses a combination of an accelerometer, gyroscope and often a magnetometer which measure linear accelerations, rotational speed and magnetic field strength respectively, imparted on an object within a local frame of reference. Some IMUs include an altimeter which detects current external air pressure giving 9 degrees of freedom plus current elevation [61,48]. For indoor environments, the use of a magnetometer to determine magnetic north, and therefore current heading, is not a viable option as local disturbances (e.g. infrastructure) can lead to heading inaccuracies of over 100° [62].

The benefits of inertial systems are the low cost and ease of scalability, due to the lack of infrastructure requirement i.e. only the IMU unit is required for asset tracking [22,63]. The reduction in cost is largely due to advancements in micro electro-mechanical

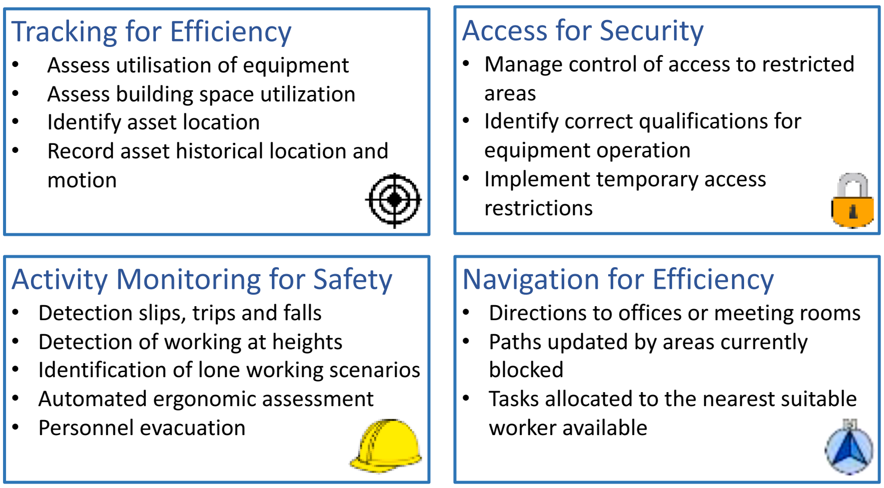


**Fig. 1.** Identified benefits of IPS and ILBS for augmented operators in the CPS.

sensors (MEMS) which has allowed the inertial sensors used in IMUs to be low-cost (e.g. LSM9DS1 a combined accelerometer, gyroscope and magnetometer) can be purchased for less than £5 each (at time of writing) [64] Critically, IMUs meet requirements for industrial/mass deployment using low-cost sensors since they can be used for estimating position without the need for adjustments to infrastructures. The main limitations are the potential accumulation of errors over time and the need for a supporting technology to enable absolute positioning values to be determined. Although IMUs can be used as standalone systems (e.g. [65,66]), MEMS sensors are prone to error due to sensor bias [13]. Examples include sensors providing a non-zero value under stationary conditions (i.e. zero non gravitational accelerations) or random sensor noise [67]. These errors are a particular issue for *strapdown* techniques because the error is magnified by the double integration methods [63], often leading to positional error (proportional to $(\text{error})^2$) quoted as a percentage of path travelled (0.3–1.5% for foot mounted solutions [22,63]). Additionally only relative location is possible rather than absolute position [68].

### 4.2. Wi-Fi networks and wireless local area networks

Wi-Fi Networks or Wireless Local Area Networks (WLAN) have been indicated for ILBS in the literature due to their increasingly ubiquitous availability as fundamental infrastructure components [22]. Additionally, the availability of Wi-Fi capability in many portable devices such as mobile phones has increased interest in developing IPSs using Wi-Fi [25]. Many of the parameters of the Wi-Fi signal can be received without the need to connect to the local network or individual node (referred to as an access point (AP)) [69] enabling location information to be available without the need to be registered within a network. However, the received signals can vary significantly between different smartphone makes and models affecting performance [25]. Several commercial applications for Wi-Fi based ILBS were identified including RADAR, HORUS and COMPASS [70,71]. Unfortunately, these methods were reported to be unreliable (due to the variability in signal strength) with quoted positional errors of 8 to 10 m, making these only suited to room level accuracy when using Wi-Fi alone [25].

Wi-Fi may be suited to environments where there are a large number of humans (e.g. highly populated areas of a factory or warehouse) because the frequency operation, in the order of GHz, is less resonant with the human body ($f_{res} = 250$ MHz) so will incur a lower specific-energy absorption rate resulting in less interference [72]. Wi-Fi traditionally operates in the 2.4 GHz band but has been extended to include the 5 GHz band in an attempt to reduce signal crowding [73]. Recent developments in the Wi-Fi protocol 802.11ax, marketed as Wi-Fi 6, reportedly operates in bands up to 7.125 GHz and bandwidths of 160 MHz [74,75]. These advancements may allow for improved accuracy using time of flight as higher clock frequencies could give cm level accuracy [59]. The key benefit to adopting Wi-Fi based solutions is that the infrastructure requirements are often in place for most buildings removing the need for additional infrastructure costs [22,71]. Accuracy is dependent on the availability of APs [14] and infrastructures with a low number of installed APs may prohibit Wi-Fi location or reduce accuracy. More advanced techniques can utilise channel state information (CSI) which describes the amplitude and phase of Wi-Fi signals, used to calculate parameters such as angle of arrival [76]. A further benefit is the general coverage of Wi-Fi, with range of around 100 m per AP and the growing number of devices with Wi-Fi capabilities [22,71].

The main drawback regarding the use of Wi-Fi for ILBS is the potential collision of multiple devices in close proximity, resulting in interference (e.g. packet collision or dropped data) which can affect signal strength [22,71]. Furthermore, Wi-Fi communication has a high-power consumption (average 216.71 mW) reducing battery life on mobile devices when used continuously to detect location [14]. Finally, the typical Wi-Fi scanning rate is too slow (3–4 s in a smartphone [14]) which results in low location refresh rate and fluctuations in signals if the user is moving within a complex / variable environment [14,71].

### 4.3. Bluetooth networks

The introduction of Bluetooth low energy (BLE) has allowed mesh connections of Bluetooth devices with low power consumption (i.e. $<$10 mW) [14]. Bluetooth signals are detected by BLE beacons which can then determine the location of the person based on signal strength (RSSI) in conjunction with beacon location. In the last few years, the Bluetooth protocol has included "In-phase and Quadrature" (IQ) sampling allowing for angle of arrival and angle of departure calculations using Bluetooth [77,78]. Bluetooth low energy is designed to be low cost and is optimised for short range communication (i.e. theoretical range $<$50 m) using wireless transmission [79]. Many smartphones and mobile devices have Bluetooth capabilities and newer models are equipped with BLE [80]. For example, the use of BLE has been successfully deployed commercially in public spaces such as airports and shopping centres[71,66]. Several studies recommend using BLE or a combination of Wi-Fi and BLE protocols to determine indoor location [81–84]. The results have been variable with [79] claiming lowest average errors of 1.58 m whilst [84] stating 90% precision within 1.5–2.5 m.

The main benefit of BLE is the very low power consumption (typically 0.367 mW) allowing battery operated devices to function for years without battery replacement [14]. In addition, many mobile devices are equipped with hardware for BLE, allowing secure and efficient communication [14,71,65]. The drawbacks include interference and unreliable detection of BLE signal through the Beacons used to detect them. The frequency of use of BLE is limited to the heavily populated 2.4 GHz band and may encounter and cause interference with Wi-Fi. Additionally, as noted in [82] that the emitted BLE signals are not always detected by Beacons and therefore may not be a reliable sensor for ILBS.

### 4.4. Wireless sensor networks

Wireless sensor networks (WSN) are an established technology, built to the industrial standard IEEE 802.15.4 which defines the data rate limits and frequency bands of Low-Rate Wireless Personal Area Networks (LR-WPAN) [85]. Like Wi-Fi and Bluetooth, WSN operate in a variety of frequency bands depending on the host country. In all countries, the WSN are limited to 250kBps, and have a typical transfer range of 1–100 m whilst maintaining low power consumption [85]. Literature is currently focused on tackling the implications of limited data rate problem for WSN [86–90]. The 2.4 GHz band is used in much of the world, however WSN operates at 868 MHz in Europe and 915 MHz in the US [71]. These bands may lead to different accuracies depending on the areas of the world the technology is used due to frequency directly relating to power loss. Systems using WSN technologies, for example using Arduino and XBee (WAX), have achieved mean accuracies of 0.83 m in a laboratory environment [91]. A common WSN technology used for localisation is ZigBee [71]. Bianchi et al. (2020) developed an IPS using WSN, focusing on received signal strength indication (RSSI) localisation using Zigbee. Their system enabled room level granularity through RSSI fingerprinting and RSSI thresholding, reporting 98% accuracy however no exact location information is determined. Furthermore, the equipment density for this system was high, using 8 sensors in a 40m$^2$ test space. Uradzinski et al. also utilised Zigbee to develop an IPS based on fingerprinting using 4 nodes in a 211 m$^2$ area and reported accuracy of 0.81 m [92]. The limitation of this testing is that it is conducted over one floor only and over a long thin corridor allowing LoS conditions. ZigBee was identified as scalable (networks can contain up to 65,535 devices) and suited to industrial indoor positioning applications where many assets need to be tracked [93].

The key benefits to WSNs are low power consumption [94–97] (typically 17.68 mW for ZigBee) and the mesh network topology for improved network range, scalability and low latency [14,71]. The main limitations of WSNs are that they are less widely adopted so hardware and software support is not as readily available. In addition, the signals are more prone to interference than other technologies because the WSN signals are transmitted at low power [14]. Further research is required within realistic scenarios with non-line-of-sight (NLoS) conditions to evaluate the effect on location accuracy and number of required nodes.

### 4.5. Radio frequency identification

Radio frequency identification (RFID) systems use an antenna connected to a reader which emits frequency pulses received by RFID tags. RFID operates at low, high or ultra-high frequencies [98]. The type of RFID suitable for the localisation task is dependent on the frequency band in which the technology operates, with increasing frequency leading to increased read range. Low Frequency (LF) RFID, known as NFC, operates at 30–300 kHz [98] with a typical commercial read range of up to 10 cm [99]. High Frequency (HF) RFID operate from 3 to 30 MHz [98] with a commercial reported read range of 10 cm to 1 m [99]. Ultra-high frequency (UHF) RFID operates from 300 MHz to 3 GHz [98] commercially quoted location distances >1 m [99]. The limitations on read range of LF and HF RFID, often leads to UHF RFID being used for location applications [100]. Jeevarathnam et al. (2018) explored the use of passive UHF RFID tags as a localisation method combining read count and received signal strength (RSS). The system robustness has been evaluated via four tests: (i) various tag orientations, (ii) nearby metal objects, (iii) one additional tag and (iv) two additional tags [101]. The authors reported accuracies of 0.29–1.49 grids, where a grid is 0.16m$^2$. However, the study is limited by the small test area (0.96 m$^2$). Mo et al. (2019) tested a planar passive RFID location system, using four ceiling mounted antennas and a grid of virtual reference tags to estimate location of an unknown tag based on interpolation of its nearest virtual tags. The system reports an accuracy of 10 cm for 90% of the tags. However, the results are valid for planar orientations and small-scale lab environment (0.5 m$^2$), not representative of real-world use cases.

RFID technologies are popular due to the low cost of passive tags [102]. The key benefits of UHF RFID are the NLoS performance and that they do not interfere with Wi-Fi or BLE signals due to utilising a different RF band [14]. A critical limitation of RFID in industrial environments is that metallic objects and liquids can greatly reduce the read range of UHF RFID tags [103] and multipath effects on the signal and resulting signal fluctuation, directly affect location estimation [14]. Furthermore, the hardware is not commonplace in many environments, nor embedded in mobile devices such as mobile phones [14]. There is also a potential risk to privacy when using RFID, particularly for passive RFID tags which lack the computational capacity for cryptography [63].

### 4.6. Ultra-wide band

Ultra-Wide Band (UWB) is another RF technique that creates a pulse covering at least 20% of the central carrier frequency, with bandwidths often greater than 500 MHz [104]. The technology allows low energy transmission containing large amounts of information [104]. UWB provides high accuracy in the presence of severe multipath effects and is not greatly impacted by the presence of walls [104]. UWB read errors can be corrected via the detection of multiple time delayed versions of signals facilitated by the wide bandwidth [17], with positioning errors reported as less than 10 cm [17,105]. These features make UWB ideal for many indoor positioning service requirements that prioritise accuracy. Very few experimental studies have been identified that evaluate the performance of UWB in realistic industrial scenarios [6]. Correa et al. suggest the benefits of UWB will become diminished with the introduction of 5 G networking which is planned to be installed ubiquitously and with similar wideband capabilities and cm level accuracy [22]. Schroeer attempt to improve UWB accuracy in industrial scenarios through the use of multichannel location, reducing the overall estimation mean from 0.22 to 0.13 m [106]. The system is still impacted by NLoS and multipath effects increasing the error by 0.2 m with the limitation that no dynamic testing was undertaken. There are conflicting reports about the impact of NLoS on UWB. In [71] UWB is reported to be unable to accommodate NLoS, or is no better than Wi-Fi and BLE [107]. However, [22,63] claim UWB yields many outliers when under NLoS conditions.

The main benefits of UWB are the improved time resolution and data rate whilst using a short wavelength making signals significantly less impacted by multipath effects or human interference [14,71]. UWB can achieve cm level location accuracy and therefore better granularity for ILBS [71,14]. The limitations of UWB are that some of the technology options are expensive. At the time of writing, 12 UWB development boards cost approximately £140 (≈£12 per unit) [108] so it may be a costly system to deploy due to the number of nodes required (i.e. 4 per room plus 1 per asset tracked) and the impact on infrastructure. From the reviewed literature, it remains unclear regarding the effects of NLoS and may require complex algorithms to identify and correct for this [22,63]. Additionally, UWB is an energy intensive technology (i.e. >1 W usage) [91] and is affected by metallic and liquid surfaces [104].

### 4.7. Acoustics

Acoustic systems require an emitter and a receiver which can be placed on the environment or the item to be tracked. Ashhar et al. (2019) used an array of slow-moving ultrasonic sensors using Doppler shift compensation to improve location estimation, reporting accuracy less than 10 mm. This work is limited however to low velocities (max 1 m/s), small distances (<4 m) and in LoS conditions [109]. In addition, the test setup was planar, so the implication of out of plane motion is unknown. Vera-Diaz et al. proposed a convolutional neural network (CNN) to estimate speaker location in three-dimensions from a microphone array achieving accuracy of 0.5–1 m however details as to the test and training accuracy of the CNN are not given [110].The accuracy of acoustic systems utilising ultrasonic sensors (e.g. > 20 kHz) are an order of magnitude more accurate than the regular acoustic (e.g. 20 Hz to 20 kHz) method. For example, a system called *The Bat*, requires the user to wear an emitter which sends ultrasound pulses to ceiling mounted receivers and reported positional errors as low as 3 cm [111]. The reverse of this, a system called *The Cricket*, uses ceiling mounted emitters with the user wearing a receiver, this method achieved positional accuracy within 5 to 25 cm [111]. The accuracy of ultrasound systems is usually reported at cm levels (quoted errors of approximately 15 cm and as low as 3 mm [22,71,109,111]), however, acoustic waves travel much slower than RF and are more likely to interfere with one another.

The main advantages of ultrasonic and sound location systems are that the hardware is inexpensive [71] and processing is relatively simple and accurate due to the significantly slower propagation speeds (when compared with RF signals) which enables time of flight techniques to be used more easily [71]. Although acoustic systems have been considered a simple effective technology for indoor positioning in [111], these systems have several limitations, for example difficulty in distinguishing which nodes sent pulses and ambient noise sensitivity [111]. The slower propagation of acoustic systems gives rise to other issues such as the Doppler effect where motion impacts the time a signal is detected [109]. A further limitation of acoustic systems includes the requirement for ceiling mounted nodes, adding additional infrastructure cost [22]. The impact of ambient noise and multipath effects can lead to erroneous or inaccurate readings with complex algorithms often required to account for this [22].

### 4.8. Computer vision

A computer vision system utilises cameras to view the area around an asset, it then determines location based on scene matching and comparison of key location markers with a known database [71]. Mobile vision systems tend to use fixed feature-based Simultaneous Localization and Mapping (SLAM) algorithms [22]. In SLAM, an environmental map is created by detecting visual features via an on-board camera and location is determined based on relative position to these features. Vision based location is commonly used in the robotics domain but has also been used for pedestrian location [71]. Jabborov and Cho developed a vision based IPS using a smartphone camera, utilising reference images captured within the location. The authors report an average location error of 0.15 m [112]. This system benefits from being non-RF (avoiding interference), not impacting infrastructure and being usable in both indoor and outdoor environments. The system however is limited as it requires distinct and unique features within the environment and, as with all vision-based systems, requires LoS and a database of high-resolution images of the reference objects. The benefit of this system is it can progressively build up a map and improve its location estimation over time as features are revisited and seen multiple times [22]. Furthermore, a vision-based system requires no prior knowledge of a location or the map of the area as this is built as the asset moves.

There are a range of limitations for vision systems which reduce their use in ILBS. Computer vision techniques are unlikely to be an effective solution for location of assets particularly in a dynamic factory or warehouse where the environment is continually changing as assets move throughout the processes [113]. It also poses significant challenges regarding the GDPR due to the nature of capturing images for analysis [114]. In addition, components can be expensive (e.g. £100 for camera modules [63]) with high computation, storage and energy requirements (e.g. up to 1 W for cameras [63]). As such vision systems are poorly suited for an inexpensive low power IPS [63]. Vision systems are also impacted by lighting conditions which may be variable in a manufacturing workspace [14] Compilation of images is a large undertaking and can require significant storage space and processing power. [105].

### 4.9. Summary of technology and recommended use cases

The data in Table 1 were extracted from [13,22,57,69,70,115,116]. Costs of hardware vary over time but are included for completeness, identified at the time of writing in [64,117–119]. Based on the literature, the authors provide the following recommendations for technology selection based on use case requirements, these recommendations are by no means exhaustive and may not be applicable to every scenario, care should be taken by the practitioner to compare the benefits and limitations of each technology presented to best suit each individual use case regarding accuracy, security, implementation, cost and maintenance requirements.

As can be seen from the table and to be expected the GSSN is not suited to any form of indoor location service where line of sight is heavily restricted yielding highly variable accuracy. Inertial systems should not be used alone and will require an updating technology due to the relative nature of inertial estimations. The impact of drift on low cost IMU units is also significant so they should only be used in scenarios where a repeated zero-velocity point occurs for example in pedestrian dead reckoning. Wi-Fi, Bluetooth and WSN can all

**Table 1**
Summary of technology approaches and corresponding accuracies for research and commercially available IPS technologies.

| System Technology | Cost (Antenna) (May 2020) | Cost (Receiver) (May 2020) | Infrastructure Impact | Privacy/ Security | Power Consumption | Coverage | Scalability | Accuracy | Impact of metal interference | Line of Sight Needed | Signal Type | Limitation |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **GNSS** | N/A | £10–£50 | None | High | > 150 mW–1.5 W | Global | High | 3 m–15 m+ | Moderate to High | Yes | Radio Frequency | Multipath impacts and signal attenuation |
| **Inertial System (MEMS)** | ≈£15–£30+ | | None | Med - High | Very Low | 10 $m^2$–2500 $m^2$+ | High | 0.8–2% of path travelled Drift can impact error by >0.5 $m/s^2$ | None | No | Environmental /Physical Phenomena | Relative location only<br>Sensor bias and noise |
| **Wi-Fi** | ≈£5–£10 | ≈£30–£150+ | High unless already existing | Med - High | > 500 mW | 10 $m^2$–2500 $m^2$+ | High | 0.4 m–5 m | Moderate | No | Radio Frequency (2.4 GHz or 5 GHz) | Ambient Interference<br>Low accuracy<br>Expensive if not installed |
| **Bluetooth LE** | ≈£5–£30 | | Moderate | Med - High | 1 mW –50 mW | 10 $m^2$–1000 $m^2$+ | Moderate | 0.9 –2 m | High | No | Radio Frequency (2.4 GHz) | Impacted by ambient interference<br>Low Accuracy |
| **WSN (Zigbee)** | ≈£5–£30+ | | Moderate | Low-Med | 17.68 mW | 10 $m^2$–100 $m^2$ | High | 0.8–5 m | Moderate | No | Radio Frequency (2.4 GHz) | Short range and high latency<br>802.15.4 protocol<br>Less used technology |
| **UHF RFID (Passive)** | ≈£100–£250 + £50–£100 per antenna | £0.01-£0.03 | Low | Low - Med | > 300 mW | 10 $m^2$–1000 $m^2$+ | Moderate | 1 m–5 m | High | No | Radio Frequency (860 MHz to 960 MHz) | High Antenna and Reader cost<br>Severe impact of metal |
| **UHF RFID (Active)** | £250+ + £50–£100 per antenna | £50–£100+ | Moderate | Med | > 250 mW for Reader | 10 $m^2$–1000 $m^2$+ | Moderate | 0.1 m–1 m | Moderate | No | Radio Frequency (860 MHz to 960 MHz) | Metal interference<br>High unit cost<br>Frequent maintenance |
| **Ultra Wideband** | ≈£10–£200 | | Low-Moderate | Med-High | >1 W | 10 $m^2$–400 $m^2$ | Low | 0.3 m–0.5 cm | Moderate | No | Radio Frequency (3.1 GHz to 10.6 GHz, 500 MHz bands) | High power consumption<br>High infrastructure impact |
| **Acoustic (Ultrasound)** | ≈£2–£20 | ≈£10–£20 | Low-Moderate | High | > 300 mW | 10 $m^2$–2500 $m^2$+ | High | 0.1 m–0.6 m | None | Yes | Ultrasonic (20 kHz) | Line of sight<br>Ambient sound interference |
| **Vision** | ≈£10–£250 | | High unless already existing | Low - High | > 1 W | ≈6 m (Resolution and LoS dependant) | Low | 1 mm –0.3 m | None | Yes | Light | Requires LoS conditions<br>Impacted by lighting conditions |

be used in a lower accuracy service (i.e. room level granularity) or for proximity detection, they also have comparable cost implications and infrastructure impact (assuming Wi-Fi already exists in a building) however for battery consumption the use of WSN of BLE is preferred where it is a viable option. Due to the range limitations the use of passive RFID is recommended as a way pointing technology only, this could therefore be used to update an inertial system at known locations or for services that need to ensure an item or person has passed several checkpoints at specified times or in a specific order (for example in a manufacturing process).

In scenarios that will have mostly LOS conditions (e.g. large open spaces or warehouses) and where high accuracy is a requirement of a system (e.g. personnel or vehicle location within a factory) Active RFID, UWB and acoustic technologies would be recommended. Active RFID benefits from additional sensors often fitted to the same device enabling collection of additional environmental data, however. UWB is better suited where some NLOS conditions may occur and has an added benefit of data transmission in addition to location estimation, however, is less scalable and more costly that an ultrasound system. Both technologies suffer from high power consumption so any services will need to incorporate battery charging for the mobile nodes and ideally mains power supply for anchors to minimise the maintenance impact.

Vision systems are a viable option for asset tracking under LOS conditions however users should be wary where personnel may be involved due to the implications of GDPR or where trade secrets or sensitive information may be visible in the event of a data breach.

## 5. Techniques

Techniques refer to the type of location algorithm used to process the data from the sensor technology. The technique will largely depend on the type of technology adopted and the techniques have been aligned to their respective technologies in Fig. 2. Inertial responses utilising IMU sensors, are analysed for dead reckoning to give relative position changes in place of absolute locations (Section 5.1). Techniques used to analyse RF technologies can be segmented into *range-based* or *range-free* approaches [22]. In *range-based* approaches, three methods have been identified: (i) *received signal strength* to determine location based on mathematical signal loss as a function of distance from an antenna [14], (Sections 5.2, 5.2.2 and 5.2.3) (ii) *time domain characteristics* such as the time taken for a signal to be sent and received, known as time of flight (Section 5.3) or (iii) *using an array of antennas* to extract angle

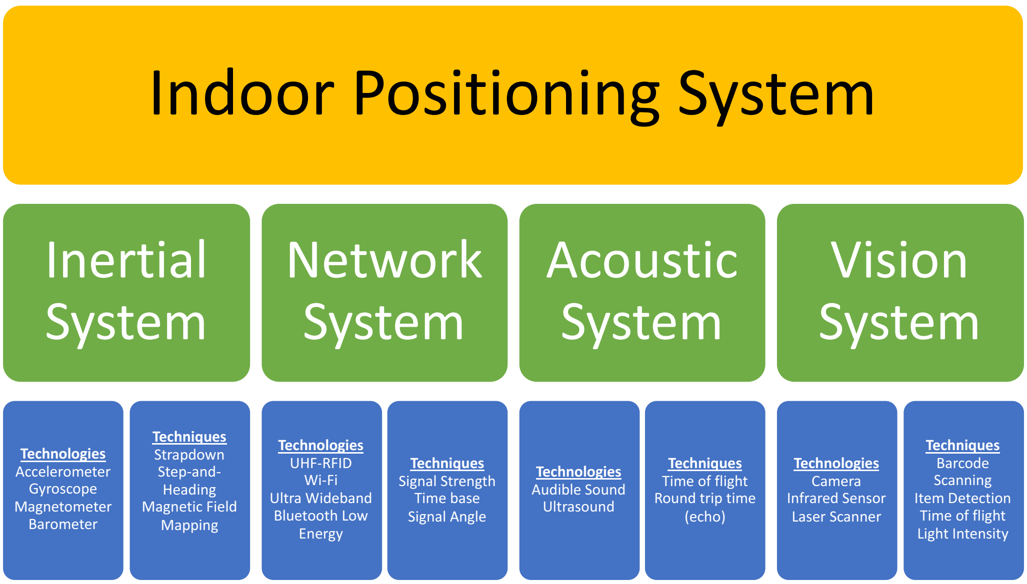


**Fig. 2.** Summary of indoor positioning technologies and techniques for each of the four broad categories. A hybrid system may utilise any combination of the technologies and techniques.

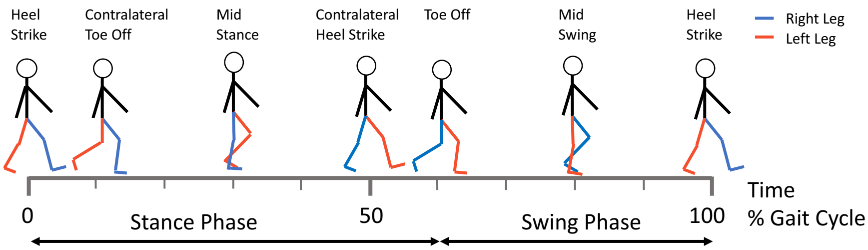


**Fig. 3.** One gait cycle with an emphasis on the right foot (blue) and left leg (red) redrawn from [122].

characteristics (e.g. angle of arrival) to estimate location [14] (Section 5.4). Using *range-free* approaches, location can be estimated based on proximity to an anchor node, typically using RSS [22]. The RSS data can be used for signal fingerprinting where unique patterns of signal strength are recorded and saved in a database and used as a lookup table to estimate location(s) [22]. Signal emitting devices can use a range of algorithms and machine learning (ML) techniques to detect features within the received signal data, enabling absolute location to be determined with respect to known emitter locations [13,24].

### 5.1. Inertial dead reckoning

Dead reckoning is a method of determining the target's *next* position based on *current* location [120]. The direction of travel (heading) and how far it has travelled in that direction [120] are used to give a relative location based on last known position. When used for human tracking, this is referred to as pedestrian dead reckoning (PDR) [62]. The use of IMU is particularly suited to human asset tracking, typically utilising either a *strapdown* or a *step and heading* technique [22,62].

#### 5.1.1. Strapdown

In the *strapdown* technique an IMU sensor is mounted with its three principal axes aligned to three orthogonal axes of the rigid body to be tracked. There must be no relative motion between these parts so that the changes in the signal measurements can be directly attributed to the motion of the asset(s) and not an artifact resulting from attachment variability [121].

Throughout the literature, the global frame of reference (i.e. the dynamic acceleration with respect to the ground) is predominantly used for the *strapdown* technique. This is calculated through subtracting gravity and integrating the signal results twice to get position changes in three-dimensional space (see [22,65,66,120]). However, it is unclear how the global reference frame is acquired and real time implications of the systems are not described. The accelerations measured are integrated twice to get distance travelled and the angular speed is integrated to give heading [22]. The calculations required for the *strapdown* technique are given in Eq. (1) (velocity from acceleration), Eq. (2) (position from acceleration) and Eq. (3) (heading from angular speed) respectively, established in [22]. Double integrations can lead to quadratic error propagation over time so inaccuracies / offsets in either acceleration or angular speed cause the location to deviate from the ground truth, a phenomenon referred to as drift [22]. Drift can be minimised through use of Zero-Velocity Updates (ZUPT) [22].

Equation 1 – Derivation of velocity from acceleration measured by an IMU [22]

$$\boldsymbol{v}(\boldsymbol{t}) = \boldsymbol{v}(0) + \int_0^t (\boldsymbol{a}(\boldsymbol{t}) - \boldsymbol{g})\boldsymbol{dt} \tag{1}$$

Equation 2 – Derivation of position from acceleration measured by an IMU [22]

$$\boldsymbol{x}(\boldsymbol{t}) = \boldsymbol{x}(0) + \int_0^t (\boldsymbol{v}(\boldsymbol{t}))\ \boldsymbol{dt} = \boldsymbol{x}(0) + \int_0^t \int (\boldsymbol{a}(\boldsymbol{t}) - \boldsymbol{g})\ \boldsymbol{dt} \tag{2}$$

Equation 3 – Derivation of heading from angular speed measured by an IMU [22]

$$\boldsymbol{m}(\boldsymbol{t}) = \boldsymbol{m}(0) + \int_0^t \boldsymbol{v}(\boldsymbol{t})\ \boldsymbol{dt} \tag{3}$$

Zero-Velocity Update is a process which leverages a sequence of events that are known to have a regular stationary point (i.e. the velocity is zero), to correct inaccuracies arising from integration of the recorded accelerations. For ILBS supporting human assets, for example for navigation, sport or safety applications, this is often achieved using a foot mounted IMU, in combination with the human gait.

Human gait motion is cyclic in nature, consisting of a *stance phase* and a *swing phase* as illustrated by Cheung et al. [122], redrawn in Fig. 3. Approximately 60% of a gait cycle consists of the *stance phase (where one foot is stationary)* and the remaining 40% consists of the *swing phase (where one foot is moving forward and the other is stationary)* [122]. During the stance phase, the foot is in contact with the ground, meaning the velocity of the foot at that point is zero. Because of the known zero position the use of gait analysis, specifically the *stance phase*, was proposed as a method for drift correction when monitoring walking motion of humans by Yun et al. [65]. Identification of the stance phase can be used to correct drift of the IMU by resetting the velocities to zero when this phase is detected [65]. In practice, ZUPT takes the offset from zero, where a stationary point is detected and divides the offset by the number of data points since the last stationary point (i.e. the previous step) yielding a drift value. This drift value is subtracted from each point in the velocity successively, giving a better representation of the real velocity.

In the literature, the ZUPT approach has been used to provide accurate location information. A foot mounted IMU inertial location system, utilising a ZUPT corrected *strapdown* approach combined with an extended Kalman filter (EKF) [123] has been used by Zhang et al. [124]. Their system is able to yield location accuracy of 0.217%, 0.35% and 0.207% of the path length in the XY, XZ and YZ planes respectively, demonstrating use of a high-quality inertial system to obtain high level of accuracy. However, this system is expensive as they utilise a non-MEMS IMU. A comparable unit, the MTi-100, costs €1499 at the time of writing [125].

#### 5.1.2. Step and heading

The *step and heading* technique uses features within the acceleration response from the IMU to detect when the wearer has taken a step (i.e. step detection) it therefore inherently can only be used for personnel tracking or assets carried by people [126]. An estimated step length is used in combination with the heading (from Eq. (3)) to determine location [22,126]. The *step and heading* distance errors are linear rather than quadratic, therefore location error propagation is reduced to two dimensions. However, stride length variation (between people and throughout walking) introduces an additional source of error [22]. To overcome this issue, the use of RF technologies is common in the step and heading technique [14,22,57,61,126–128]. Several researchers utilise the on-board sensors in smart phones for location detection services. Zhang et al. (2021) utilised the phone Wi-Fi and inertial signals to improve location estimation for pedestrian tracking in conjunction with step and heading [127]. They assessed three approaches for combining the data: (i) a Kalman filter (KF) (ii) a Particle Filter (PF) and (iii) a long short-term memory (LSTM) neural network. The best result was achieved from using the LSTM algorithm, compared with the PF and KF (i.e. an average error of 0.42 m, 2.69 m and 3.32 m respectively) [127].

Although step and heading is used in the literature, the precise algorithms are not always detailed. A step detection solution supported by BLE beacons was reported by Chen et al. [128]. Their study compared step detection when considering various positions of a portable phone: (i) whilst texting (phone in hand, mid body), (ii) swinging (phone in hand to side of body), (iii) walking whilst making a phone call (phone to ear) and (iv) in the users' side pocket (on hip) [128]. Their system determines whether triangulation is possible by checking if the beacon count was visible (i.e. three or more beacons detected), otherwise a fingerprinting approach is used (see section 5.2.1) with a reported accuracy of 1–1.5 m [128]. However, it was not stated if the system was able to function in real-time or utilising historical data only. Finally, Chen et al. (2016) developed a service for the use of an IMU with time of arrival calculations from ultrasound pulses in addition to comparison with a maximum a posteriori (MAP) algorithm in which the most likely next location is determined given the previous state [61]. Their proposed MAP algorithm was shown to outperform an EKF and a PF. In a comparison study they reported a reduced mean error from 0.98 m (when using an IMU alone) to 0.31 m over a path >50 m within a room 25 m by 8 m (when used in conjunction with ultrasound) [61].

### 5.2. Received signal strength approaches

#### 5.2.1. Received signal strength and fingerprinting

Variations in building structure cause large differences in signals due to NLoS, shadowing and multipath propagation resulting in an apparent uniqueness of received radio signals such as Wi-Fi, Bluetooth or WSN from different positions around a building. Received Signal Strength Fingerprinting, also known as radio mapping, utilises these effects of building complexity on RF signals as a means of identification of location. The RSS measured at different locations is used to identify areas in a building with the same or similar signal patterns to identify position either deterministically or probabilistically [22]. The fingerprinting process can be applied to any RF signal as demonstrated by Liu et al., where Wi-Fi signal fingerprinting is analysed for location using k-means clustering [129]. Jain et al. used BLE signals for fingerprinting to compare analysis algorithms. The random forest machine learning algorithm was the best for extracting the location estimation although location error was not reported. Finally, Peng et al. used passive UHF RFID tag for RSS fingerprinting. They analysed results using a deep convoluted neural network to estimate location reporting a maximum location error (MSE) of 1.432 m [130]. The use of RFID fingerprinting is less common than Wi-Fi and BLE signal fingerprinting due to the latter technologies being readily available through smartphones [69].

Although RSS distance measurements and fingerprinting are the least complex method the main limitation of fingerprinting approaches are that a large number of calibrated points are required for location accuracy, resulting in arduous and often manual collection of calibration measurements and longer time scales to obtain fine granularity [15,131]. Researchers have attempted to address this issue through reducing the burden of calibrating of the system using automation and crowdsourcing approaches [15,131,132]. However, fingerprinting setup remains an expensive and time-consuming process which may require iterative, ongoing calibration with data storage adaptation as event patterns, occupancy, equipment or infrastructure changes over time [133]. Additionally, this technique is not suitable for indoor location in a dynamic manufacturing space where large, often metallic, components may be frequently transported around a factory floor i.e. changing the location fingerprint.

#### 5.2.2. Proximity

Proximity location techniques use known RF emitter nodes to detect when the RSS of the receiver is above a threshold, i.e. the receiver is within range of the emitter. When multiple sensors are recorded above the threshold, the node with the highest RSS is considered the closest node [134]. Effectively, this is an RSS or RSSI ranging method with coarse boundaries to allow the measurements to be more stable [135]. Threshold boundaries can be set to locate assets within predefined *immediate, near* and *far* vicinities.

Examples of ILBS requiring room level accuracy are ambient assisted living and targeted marketing [136]. Jiménez et al. used proximity of a wrist worn receiver to BLE beacons, in addition to binary sensors, to estimate an approximate location of a user. They reported an accuracy of 1.5 m for 80% of the movement over 10 days, using capacitive floor sensors as the location ground truth [137]. Muddinagiri et al. (2020) used proximity of BLE beacons as part of an application for marketing furniture to customers in a store, however no specific information was given on accuracy of this system and the location was not designed to be exact.

#### 5.2.3. Received signal strength or received signal strength indication ranging

Techniques utilising RSS or RSSI use reference tags or beacons attached to known locations. A receiver is used as the unknown location to measure the RSS, often reported in decibel-milliwatts (dBm) where 0 dBm is 1 mW, or RSSI (a unitless 8-bit quantised value) [138]. The recorded RSS value is then matched to a signal propagation model to estimate the distance from the known anchors in range.

A common approach is to use a standard power loss model, where dB is proportional to logarithm of distance [22]. To estimate the location based on distance, trilateration is used, requiring a minimum of 3 or 4 nodes to determine location in 2D and 3D respectively. Several studies use RSSI as a variable for location, with accuracy reported ranging from general room level granularity to location accuracy within 5 m [9,139,140]. The variability in RSSI or reduction of signal strength (e.g. by 3–12 dB [141]) has been reported due to obstructions of the antenna [141]. In addition, RF signals suffer from multipath propagation and anisotropic interference, affecting location estimation [22,141]. A further source of error stems from the use of a theoretical power loss model whilst real world signals may vary from this model [22,141].

The RSS ranging technique is used with various technologies with different reported accuracies. Choi and Choi use Wi-Fi ranging to support human tracking for PDR, reporting an average 1.38 m error in ranging distance estimation [142]. However, the authors do not accurately describe the environment nor account for the interference that may be present. Venkatesh et al. use BLE ranging to estimate distance from an anchor node. They report the use of median RSS (maximum error 0.91 m at 7.5 m) as more accurate than mean RSS (maximum error 1.08 m at 7.5 m), [143]. However, they do not consider the maximum or minimum value received and do not explore how the signal characteristics influence mean or median values in real-time. Furthermore, their reported methodology uses a coarse selection of locations with 5 unevenly spaced points between 1 m and 7.5 m, and do not report what a near-zero distance would be recorded as. Gui et al. used passive UHF RFID ranging with a reference free schema to determine relative positions between tags and derive absolute locations based on reference tags [144]. The system in [144] reports location errors of 2.7 to 4.1 cm which is unusually small for an RSS ranging technique, typically in the region of 15–50 cm [63]. However the system only considers X and Y axis errors which are in the plane parallel to the antenna and within a small test area 38 cm × 28 cm. An RFID ranging approach was reported by Ma et al. , where four RFID readers, positioned to form a square, scan for passive RFID tags and record the RSS values [145]. The results have been analysed using weighted least squares, multidimensional scaling and weighted multidimensional scaling (WMDS) algorithms to extract tag distance, with trilateration used to estimate position [145]. The WMDS algorithm outperformed the other techniques, yielding a mean location error of 0.56 m. However, it should be noted the results are based on simulated data [145].

### 5.3. *Time of arrival and time difference of arrival*

Time of Arrival (ToA) systems measure the propagation time of a signal between the transmitter and receiver to compute the distance from a particular transmitter. For this technique, all nodes need to be synchronised to a single reference time which can be difficult or impossible for simple devices with limited processing power to support complex synchronisation algorithms [22]. The signal transmitted by nodes with a known location are *sent* at a given time stamp, a second timestamp is attributed to the *received* time. The difference between the timestamps is calculated as the journey time and locus distance calculated. The resulting distance can be combined with other transmitter responses in a trilateration technique to identify position. The main drawback of this technique is the requirement for accurately synchronised and high-resolution clocks. All time-based techniques suffer from processing offsets, multipath effects and NLoS resulting in an overestimate of signal travel time and therefore distance overestimate, for example an error of 1ms can lead to a distance error of 300 m [146].

To reduce the need for synchronised clocks, time-based systems often utilise Time Difference of Arrival (TDoA) which only requires clock synchronisation of the transmitters [146]. There are two approaches to achieve TDoA. Firstly, the received signal can be measured from two separate receivers and the difference calculated, reducing the importance of synchronisation [22]. Secondly, a single transmitter produces two signals with different propagation times (e.g. RF and Sound) removing the need for synchronisation between sensors but requiring additional hardware [22]. The increased complexity of NLoS detection algorithms can increase latency and reduce suitability for simple devices, which typically have limited processing capabilities [22].

A third time-based technique is known as the round-trip time (RTT) a two-way measurement where the receiver responds to a transmitter with its internal received and retransmit time. The difference can be leveraged by the transmitter to determine the RTT without the need for a synchronised clock. ToF and TDoA are approaches commonly used in UWB systems and is commercially available from suppliers (e.g. the Quorvo DWM1000 module) [147] stating location accuracy of 10 cm. Wi-Fi and BLE are less frequently used with ToF approaches due to a lack of protocol support. Wi-Fi was unable to support time-based algorithms until recently with the updated 802.11mc fine-time measurement (FTM) protocol. Hashem et al. leveraged FTM to develop WiNar, a process that uses the RTT of data packets combined with Wi-Fi fingerprinting to improve location estimation [146], reporting an average location error of 0.77 m. Giovenelli et al. proposed a hybrid BLE RTT approach supported by RSSI ranging to achieve a reported accuracy of 2.0–2.7 m [148]. The accuracy is not significantly improved by the RTT as the key limitation in this system is the internal clock speed, 16 MHz giving resolution of 62.5 ns, or a distance resolution of 9.375 m. Although they aimed to reduce this error through Kalman filtering with random white noise, it could not be completely removed, the authors state that the error can be "effectively removed" using a low pass filter but do not provide details [148].

### 5.4. *Angle of arrival and angle of departure*

Angle of Arrival (AoA) and Angle of Departure (AoD) algorithms measure the incidence angle between two nodes and require an array of antennas [78,149]. AoA uses a signal from the transmitter node and the array of antennas at the receiver calculates the incidence angle, based on the phase difference in the IQ, received by the antennas [77, 78]. For AoD the transmitter has an array of antennas and the receiver takes a sample of the IQ, received from the antenna array and calculates AoD [77]. There are two methods to determine location using the incidence angle between sensors two methods. The first uses angle and estimated distance from either the RSS or ToF/TDoA approaches [77]. The second requires a more sophisticated antenna array and two angles from one antenna array to obtain 2D location, or three angles from two antenna arrays to obtain 3D location. The incidence angle between the known sensors and the unknown node is used to create a triangle to determine the location [22,77,78]. Based upon the incidence angles, triangulation is used with each permutation of angles to determine the location of the unknown node [149]. Both AoA and AoD can yield accurate location estimates, with Bluetooth SIG suggesting an accuracy of <10 cm with 86% reliability [77].

Angle of arrival has been reported as suitable for use with several RF technologies because accurate clocks are not required, overcoming the limitation of time of flight techniques. Yang et al., combined AoA approach with a Wi-Fi CSI to estimate AoA of Wi-Fi packets [76]. Their system is reported as achieving angular accuracy of 5.82° using a two-dimensional multiple packets-based matrix pencil (2D M-MP), which produced a maximum location accuracy of 0.66 m [76]. Wang et al. utilised an AoA approach using a single BLE anchor containing an antenna array. Their system achieved an azimuth angular error of 2.4° to 2.75° and elevation angle errors of 4.8° to 5°, resulting in median positional errors of 30–36 cm after filtering [150]. The literature suggests this technique in combination with BLE is promising for future applications [150]. Finally, Lopez et al. estimated location of a receiver using an RSSI approach to AoA, with two pairs of directional RFID antennas. They estimate the angle of arrival based on the axial RSS sensitivity of the antennas [151]. Each pair of antennas determined the angle of the tag from the perpendicular antenna pair with triangulation to estimate location. This approach achieves an accuracy of 1.15 m with 90% certainty covering a 12 $m^2$ area [151]. The introduction of AoA support in the BLE protocol specification 5.1 [78] has been explored for research purposes to obtain low power usage with high accuracy using BLE devices.

## 6. Summary and gaps

The ILBS literature has been focused on characteristics of the system such as positioning accuracy, deployment costs or the computational complexity [14,22,152]. The reviewed literature does not compare the quality of the sensors that have been used to ensure they were fit for purpose (i.e. indoor or industrial operation). Furthermore, no direct comparison has been found with regards to ranging accuracy of the different RF technologies in identical or comparable experimental conditions. The reviewed research has not attempted to balance multiple requirements of a tracking system simultaneously and to identify the impact of different priorities, especially in complex environments. A lack of development standardisation has been identified as a limiting factor in many ILBS and IPS systems [14,22,63,71,107,152–154] and a gap in the literature exists in that no end to end ILBS have been developed.

Several reviews of IPS have been identified that highlight the key technologies required, each reporting that there is no "*one size fits all*" approach to enabling indoor localisation [14,22,63,71,107,152–154]. The most commonly used technology approach for accurate IPS were hybrid systems, utilising an inertial dead reckoning system supported by an RF technology [61,126,132,137,155]. The use of UWB with ToA or TDoA has emerged as an accurate IPS solution without the need for a supporting technology, able to achieve decimetre accuracy [6,17,105,156], with commercial devices identified which leverage this [147]. The installation of a UWB system, however, would be large scale (e.g. installation and permanent fixings of the units), impact on the infrastructure and may be prohibitively disruptive and costly to a smaller company.

To provide improved support for deployment and selection of IPS systems, greater rigour is required in research conducting comparisons and analysis of IPS systems. To enable fair comparison, a standardised test methodology should be used in place of comparing results obtained in different literature, considering the technology, techniques and operating environment. To develop robust systems that could be utilised for the benefit of real-world supply chains, it is necessary to develop and test systems within a real-world or representative environments, e.g. transport of assets that can be queried in real time. Although researchers are using smart/augmented PPE for improved safety, there is no consensus on the best approach to incorporate these into working practices which remains an active and open area of research. Researchers have also explored the use of augmented PPE for incident detection although the systems are limited by inadequate location information where infrastructure impact is to be kept minimal or is impractical to install. Monitoring personnel within CPS needs further attention to address safety, GDPR and accuracy. Further work is required to test systems in realistic and real-world scenarios to identify the impact of dynamic and changing environments, human activity and user influence on the appropriate function of tracking technology. The reviewed literature has tested ILBS without considering how the system would be used securely and how it would protect the privacy of its users [152]. With the introduction of the GDPR and DPA regulations this is an area of importance and must be included in the design of emerging systems. Consideration for how data are processed, stored and interacted with securely is therefore paramount to allow ILBS to be a viable real-world technology and to improve user acceptance. An ILBS should support GDPR requirements as a primary consideration when developing a service architecture. A comparison of security methods and requirement satisfaction must be considered when selecting protocols and processing data, particularly when human tracking is involved.

In both the *strapdown* and *step and heading* approaches, the focus of the reviewed literature has been to improve the accuracy of inertial systems using RF techniques for accurate and absolute location detection. However, the authors do not determine the impact of the different hardware on the capabilities of the system, nor have they explored calibration techniques that may be used to minimise errors further. Because of the variability of performance within industrial environments, additional work is required to compare the supporting RF technologies in conjunction with the reported analysis approaches. Finally, it is unclear from the reviewed literature whether the solution / proposed systems were working in real time. Due to the use of location systems to support safety and security services, accurate and real-time to near-real-time location information is paramount. Additional work is required to determine the latency of each approach and to identify implications of a real-time or near-real-time system for recording, calculating and reporting location.

Due to the scale of research in the field of indoor positioning systems and indoor location this paper is not an exhaustive review of the current state of the art. In addition, due to the veracity and velocity of papers being published in this area, this paper only represents a snapshot of the current state of the art which is ever evolving. This paper's contribution is that the key areas of research are highlighted in addition to key benefits and limitations of the technology selection for new and upcoming research to investigate. Furthermore, this paper identifies the key exploitation of indoor location for industrial applications, highlighting the need for research to migrate from IPS to IPS as a service. This paper will support technology selection and may reduce the pool of technologies based on the use case aligned with the information presented in Table 1.

## 7. Conclusion and further work

The review conducted has highlighted the key application areas, technologies and techniques presented in current IPS literature. The technologies have been summarised to support the selection of IPS for services built atop the IPS and recommendations have been suggested, identifying which technologies and techniques would be best suited based on use case. The techniques applicable to the technologies identified are then explored, highlighting the accuracy and limitations of each technique regarding complexity. Further work will focus on developing a unified framework that can support practitioners in developing ILBS to bridge the current literature gap of processing, storing and accessing data in addition to presenting this to the user through the use of an interface. Further reviews will also need to include upcoming technologies such as WiFi-7, 5 G millimetre wave and Bluetooth 5.3 as they mature.

## Declaration of Competing Interest

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